NAMAN BAREJA

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EDUCATION

Columbia University *Dec 2025*
M.S. Data Science, GPA: 3.63/4.0 | **President**, DSI Student Council | TA, Deep Learning for NLP | Coursework: Statistical Inference & Modeling, ML for Finance, ML for Functional Genomics

Delhi Technological University *May 2023*
B.Tech. Mathematics & Computing, GPA: 8.92/10 (top 5% of cohort)

TECHNICAL SKILLS

Languages: Python, R, SQL, SAS, MATLAB
ML & AI: scikit-learn, XGBoost, PyTorch, TensorFlow, Hugging Face Transformers, YOLO, Mask R-CNN, NLTK, Gensim, spaCy, Pandas, NumPy
Tools: AWS (S3, SageMaker), BigQuery, Tableau, Power BI

EXPERIENCE

Cheng Lab, Columbia Stem Cell Initiative, Columbia University Medical Centre *Feb 2026 – Present*
Data Analyst & Research Aide

- Own end-to-end ML pipelines for multi-omics integration (genomics, transcriptomics, metabolomics) in nutrigenomics research; apply dimensionality reduction and statistical modeling to identify molecular signatures and diet-gene interaction patterns relevant to target discovery.
- Conduct pathway analysis and contribute to experimental design and wet lab execution; independently drive studies from hypothesis through analysis, significantly accelerating research timelines in a translational stem cell environment.

Mailman School of Public Health, Columbia University — Dept. of Epidemiology *May 2025 – Dec 2025*
Machine Learning Research Assistant

- Built hierarchical ensemble biomedical NLP pipeline combining VADER lexicon scoring with fine-tuned BERT-based transformers to classify substance-use phenotypes in large-scale text data — methodology transferable to clinical notes, EHR mining, and literature-based drug discovery.
- Improved classification accuracy by 23% over baselines via sentence-level contextual aggregation; collaborated with epidemiologists on annotation schema and domain-expert label validation.

Bendesky Lab, Zuckerman Institute, Columbia University *Sep 2025 – Dec 2025*
Research Assistant — Computational Neuroscience

- Engineered YOLOv8 computer vision pipeline to detect and track real vs. 3D-printed fish frame-by-frame at scale; trained multi-label action classifier (biting, flaring) to quantify aggression across hundreds of experimental video hours.

Microsoft *Mar 2025 – May 2025*
CX Research Intern

- Analyzed internal documentation and industry benchmarks to map AI applications in customer research and cybersecurity; proposed AI-driven evaluation metrics and investment recommendations in biweekly reports to senior cross-functional stakeholders.

EXL Services *Nov 2023 – Aug 2024*
Business Data Analyst

- Re-engineered credit risk model pipelines (CCAR, CECL) in SQL, improving data integrity across 500K+ financial records; automated SAS pipelines reducing processing time by 15% and minimizing manual errors.
- Improved Probability of Default (PD) model accuracy by 10% via feature diagnostics and statistical analysis; validated and deployed enhancements with model owners, presenting findings to senior leadership.

IFFCO Tokio General Insurance Co. Ltd. *May 2022 – Jul 2022*
Machine Learning Engineer Intern

- Built Mask R-CNN + InceptionV3 computer vision pipeline for automated damage detection and repair cost estimation in car insurance claims, cutting adjudication time from 3 weeks to 3 days for low-value claims.
- Developed XGBoost + regression customer retention model improving retention by 7%.

Centre for Artificial Intelligence and Robotics, DRDO *May 2021 – Sep 2021*
Student Research Trainee

- Extended semi-supervised Snowball model for bidirectional relation extraction from 10K+ sentences (NLTK, Gensim); boosted F1-score 15% over unsupervised baselines via ensemble modeling and successive-halving hyperparameter tuning; presented at NCETET 2023.

PROJECTS

Predictive Analysis of Poor Mental Health Prevalence Across Neighborhoods *Sep 2024 – Dec 2024*

- Led team of 5 to model neighborhood-level mental health prevalence using demographic/environmental data; built XGBoost + scikit-learn models (78% F1-score); identified 10+ key SDOH predictors to inform targeted public health interventions.